

Title: ASR Benchmarking for Conversational Hinglish Locality Recognition

Table Of Contents:

1. Introduction
 2. Tech Stack & Models
 3. Results
 4. Tech Stack, Libraries, Tools, and APIs Used
 5. Problems Faced and How They Were Solved
 6. Evaluation Metrics
-

1. Introduction:

A custom dataset of 20 conversational Hinglish audio recordings containing Bangalore locality names was created under multiple conditions including quiet indoor recordings, fan noise, fast speech, and traffic noise. Deepgram Nova-2 was used as the baseline ASR system and compared against Faster-Whisper Small. The benchmark evaluated transcription quality, locality entity recognition, and latency using WER, CER, and inference time measurements.

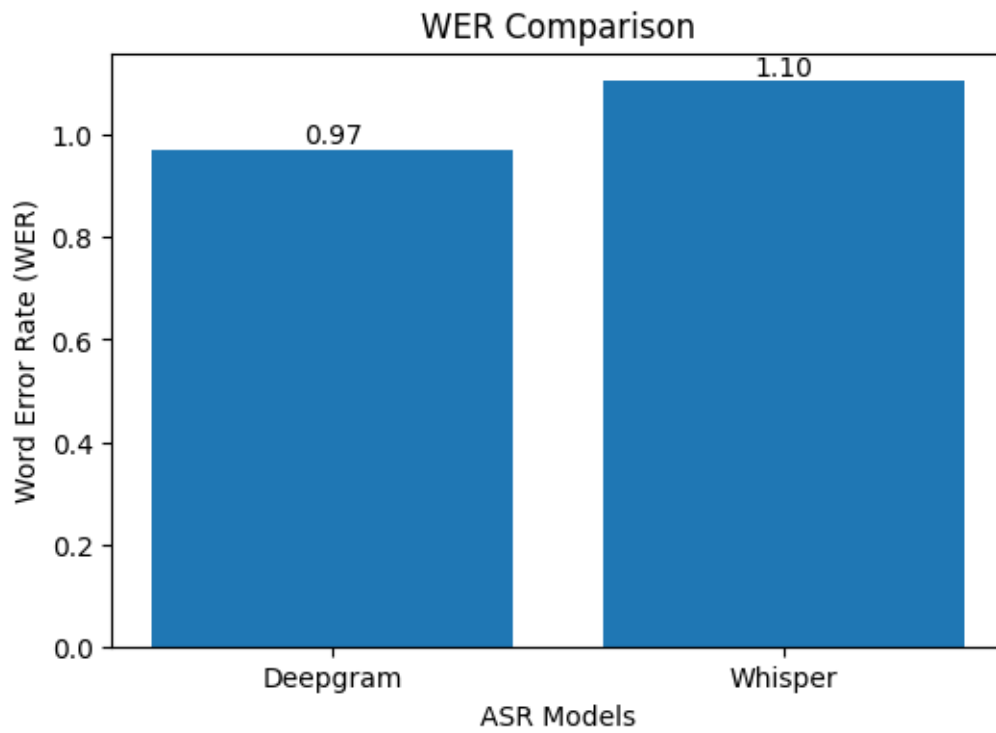
2. Tech Stack & Models:

Category	Tools/Models
Language	Python
Environment	Google Colab
Libraries	pandas, jiwer, matplotlib, faster-whisper, deepgram-sdk
Baseline ASR	Deepgram Nova-2
Comparison ASR	Faster-Whisper Small

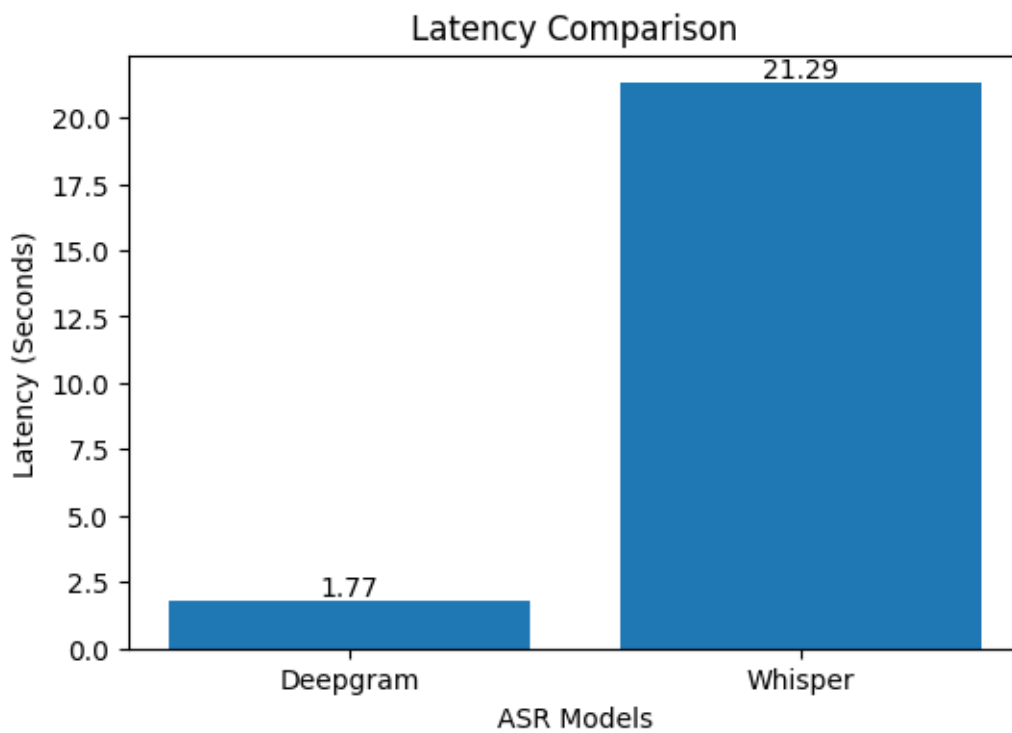
3. Results:

Model	WER	CER	Entity Accuracy	Latency
Deepgram	0.97	0.73	0%	1.76 sec
Whisper	1.10	0.89	0%	21.28 sec

WER Chart:



Latency Chart:



4. Key Findings:

Use concise bullets:

- Deepgram achieved lower WER and significantly lower latency.
- Whisper showed better qualitative multilingual robustness in some conversational samples.
- Traffic and fast speech degraded ASR quality for both systems.
- Both systems struggled with Bangalore locality recognition.

5. Failure Analysis:

Actual	Predicted
Silk Board	Silver
Koramangala	कौरवा
Whitefield	weightfield

The benchmark revealed several ASR failure patterns including phonetic confusion, transliteration mismatch, and noisy speech degradation. Both models struggled with Bangalore locality recognition under conversational Hinglish conditions. Entity-heavy utterances were particularly challenging under traffic and fast speech recordings.

6. Recommendation:

Deepgram achieved lower WER, CER, and significantly lower latency compared to Whisper on the collected conversational Hinglish dataset. However, Whisper demonstrated better qualitative multilingual robustness in certain Bangalore locality examples. Both systems struggled with Bangalore locality entity recognition due to transliteration mismatch, noisy conditions, and phonetic ambiguity. Deepgram may be preferable for production systems requiring low latency, while Whisper can be useful for multilingual conversational robustness.